Environment and Preferences

A two-period OLG economy: Y_t young and O_t old share a fixed housing stock $\bar{H}$.

- Entrants draw (y_i^y, b_i, y_i^o) from F: young income, liquid wealth, and old income.
- Young choose consumption, housing, completed fertility, saving, and lifetime tenure.
- Old choose consumption, housing, and an estate; future income is known.

Flow utilities are:

$$u_t^y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n,$$

$$u^o(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e.$$

Each child needs χ goods and κ housing; $n > 0$ is continuous. All utility weights are positive. Households discount future utility by β.

Owning adds $\xi_i \sim \text{Logistic}(\bar{\xi}, \sigma_\xi)$, independent of endowments. The warm-glow estate e does not determine entrant wealth.

Young Households

Goods and bonds trade externally at bond price $q \in (0,1)$. Housing costs P_t; mortgages finance at most share $\phi_t \in (0,1)$; the property-tax rate is τ^p.

Taking prices and the property-tax rebate T_t as given:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^y(c,h,n) + \beta V_{t+1}^R(a';i)$$

$$c + qa' + qr_t h = y_i^y + b_i + T_t, \qquad a' \geq 0, \quad h \leq h_R^{\max},$$

$$W_t^O(i) = \max_{c,h,n,a'} u_t^y(c,h,n) + \beta V_{t+1}^O(a',h;i)$$

$$c + qa' + (1 + q\tau^p)P_t h = y_i^y + b_i + T_t,$$

$$qa' + \phi_t P_t h \geq 0, \qquad h \leq h_O^{\max}.$$

Here a' is next-period net financial wealth, including mortgage repayment. Rent r_t is paid at period end; $h_R^{\max} < h_O^{\max}$.

The equal property-tax rebate T_t, current income, and wealth are available before purchase.

Old Households and Tenure

An old household has net financial wealth a and income y_i^o; an owner also holds title to H units. Taking prices and rebates as given:

$$
\begin{aligned}
V_t^R(a;i) &= \max_{c^2,h^2,e} u^o(c^2,h^2,e) \\
&\qquad c^2 + qe + qr_t h^2 = a + y_i^o + T_t, \qquad h^2 \le h_R^{\max}, \\
V_t^O(a,H;i) &= \max_{c^2,h^2,e} u^o(c^2,h^2,e) \\
&\qquad c^2 + qe + qr_t h^2 = a + P_t H + y_i^o + T_t, \\
&\qquad h^2 \le h_O^{\max}, \qquad e \ge P_{t+1} h^2.
\end{aligned}
$$

Financial saving $a^e \ge 0$ gives $e = q^{-1}a^e$ for renters and $e = q^{-1}a^e + P_{t+1}h^2$ for owners. Owners may buy or sell within $h_O^{\max}$.

Young own if $W_t^O(i) + \xi_i \ge W_t^R(i)$; this determines the owner probability $\pi_t^O(i)$. Tenure remains fixed thereafter.

Competitive Equilibrium

Given policies, F, and (Y_0, O_0, G_0), an equilibrium is a sequence of prices (P_t, r_t), rebates T_t, household choices, cohort sizes, and old distributions satisfying:

1. Young and old optimize at equilibrium prices; tenure follows the value comparison.
2. Rental entry: $qr_t = (1 + q\tau^p)P_t - qP_{t+1} > 0$.
3. Housing clears and property-tax revenue is rebated equally:

$$Y_t \bar{h}_t^y + O_t \bar{h}_t^o = \bar{H}, \qquad (Y_t + O_t)T_t = q\tau^p P_t \bar{H}.$$

4. Cohorts follow $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$.
5. G_{t+1} is generated by young choices of (a', h) and tenure, together with y_i^o.

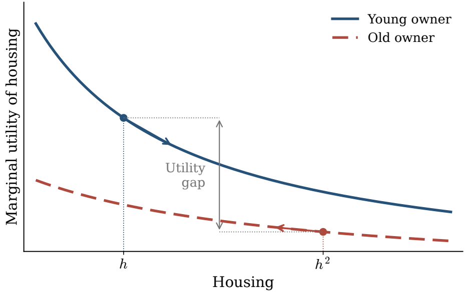
Bars are cross-sectional averages over endowments and tenure; ν converts children to entrants.

G_t is the old distribution of financial wealth, inherited housing, income, and tenure.

A positive steady state has constant prices, choices, distributions and cohort sizes, with $Y^* = O^*$ and replacement fertility $\bar{n}^* = 1/\nu$.

Housing Allocation and Welfare

At a stationary equilibrium, give each living household equal weight on its remaining utility.



With $s = h - \kappa n$, marginal housing utility is higher for a constrained young owner than for its stationary old counterpart:

$$\frac{\alpha}{s} > \frac{\gamma}{h^2}.$$

Utilitarian gain. Move a little housing from old to young, holding consumption, fertility, tenure, estates, and future allocations fixed. Old housing utility falls; total welfare rises.

Requires $\beta \geq q$, a positive share of owners with strictly binding mortgages, slack young and old housing caps, and positive old financial saving. The direct planner may relax finance.

Housing and Fertility

At fixed tenure, prices, and other transfers, a current cash gift raises fertility. More space lowers the housing cost of children; consumption also matters.

For two chosen bundles at the same ϑ, if (c_1, h_1) can accommodate n_0:

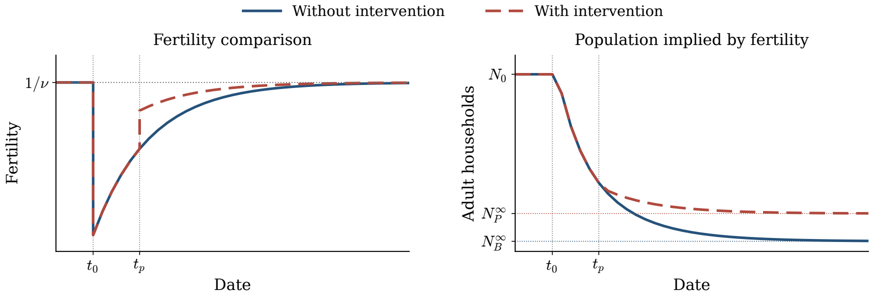
$$n_1 \geq n_0 \quad \Longleftrightarrow \quad \frac{\chi}{c_1 - \chi n_0} + \frac{\alpha\kappa}{h_1 - \kappa n_0} \leq \frac{\chi}{c_0 - \chi n_0} + \frac{\alpha\kappa}{h_0 - \kappa n_0}.$$

A rise in both consumption and housing is sufficient; otherwise their effects must be compared.

The current and old-age grants in the appendix raise treated owners' fertility at fixed prices and rebates. Aggregate effects also depend on tenure choices, housing-market clearing, and future cohort sizes.

Fertility and Population Paths

An exogenous fall in ϑ_t at t_0 starts demographic adjustment. At t_p, a housing policy branches from the same inherited state as the baseline.



Higher fertility along the policy path makes later cohorts larger: $Y_{t+1} = \nu \bar{n}_t Y_t$.

Fertility paths are assumed. Both approach replacement and yield positive population limits by cohort accounting; equilibrium housing and fiscal conditions remain to be established.